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Department of Physics, Astronomy and Mathematics

MSc Data Science

Project report: 7PAM2002 - Data Science Project

Unsupervised Customer Segmentation Using Transactional Data and Clustering Techniques

By

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Declaration Statement

This document is presented as part of the requirements for the Master of Science in Data Science at the University of Hertfordshire and is the result of my own independent work.

All external sources have been clearly cited, and I allow this report to be used for academic purposes with proper attribution.

Acknowledgement

This work represents the time, commitment, and continuous learning I invested throughout my academic journey.

I am grateful to my supervisor, Carolyn Devereux, and the University of Hertfordshire for their guidance and support in completing this project.

Abstract

This project explores how unsupervised machine learning can be used to group customers by analyzing their shopping records. For any business, grouping customers effectively is vital to creating better ads, keeping shoppers loyal, and growing total profits. The study uses a specific dataset from a UK online store that tracks one full year of sales.

A clear step-by-step process was followed, starting with data cleaning to fix missing information, delete errors, and make sure all records were accurate. We also performed a basic analysis of the data to find common shopping habits. To understand each shopper better, we used the RFM method, which tracks how recently a person bought something, how often they shop, and how much money they spend.

We tested several grouping methods, such as K-Means and DBSCAN, to see which worked best. To check the quality of these groups, we used mathematical tests like the Silhouette Score and the Davies–Bouldin Index. We also tried removing unusual data points and simplifying the information using PCA to make the results even more accurate.

The results showed that using PCA together with K-Means was the best way to create clear groups. The final version of the model found three specific types of customers: loyal high-spenders, average shoppers, and those who might stop buying soon. These groups provide useful information that can help a business make smarter choices and create personalized marketing plans..

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1 Introduction

In the modern landscape of digital commerce, businesses produce massive quantities of transactional records that track every detail of customer engagement. However, the importance of this information is found in the ability to derive useful insights that guide high-level business choices, rather than just the amount of data stored. A vital part of this analysis is customer segmentation, a process that allows companies to categorize their audience by shared behaviors to create more effective marketing plans (Wedel and Kamakura, 2012). Historically, these groups were formed using simple facts like age or basic rules, but these older methods often miss the complicated nature of how people shop in real life. With the growth of artificial intelligence, data-focused methods have become a superior choice, as they allow unique patterns to be found directly within the data without relying on old guesses or biased assumptions. Specifically, unsupervised learning tools like clustering have become common for finding invisible connections in datasets that do not have pre-existing labels (Jain, 2010), as illustrated in Figure 1.

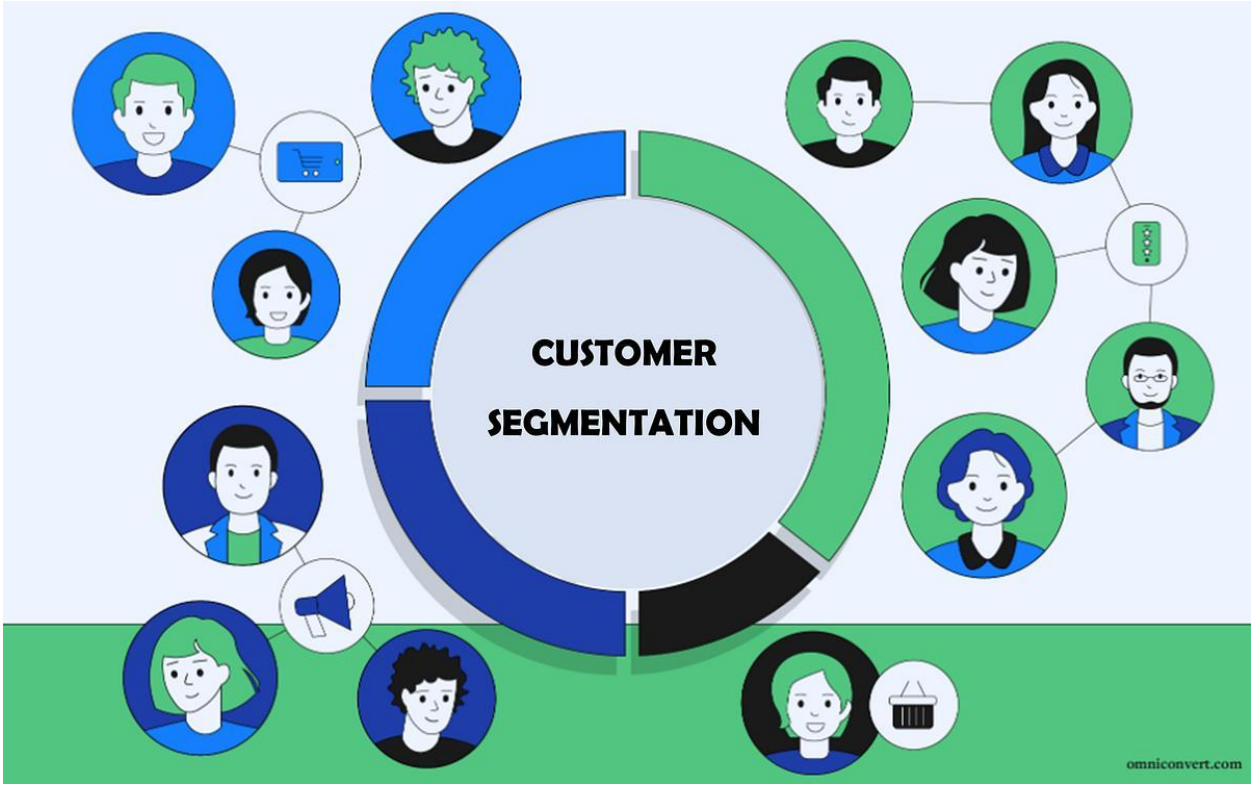

Figure 1: Customer Segmentation Concept, Source: Maleesha De Silva (2023)

One of the most popular ways to model these behaviors is the Recency, Frequency, and Monetary (RFM) system. This approach measures value by looking at how recently a customer bought something, how often they return to shop, and how much total money they spend,

offering a clear and effective way to view customer habits (Hughes, 1994). When mixed with clustering tools, the RFM system identifies specific groups that show different levels of loyalty and profit potential. This concept is visually represented in Figure 2. This research uses these advanced learning methods to group customers by analyzing a specific online retail dataset. This data includes one year of real sales records from a company based in the United Kingdom and contains both numbers and descriptive details. Before the analysis could begin, a large amount of work went into cleaning the data, which involved fixing missing records, deleting errors, and building specific features for each customer.

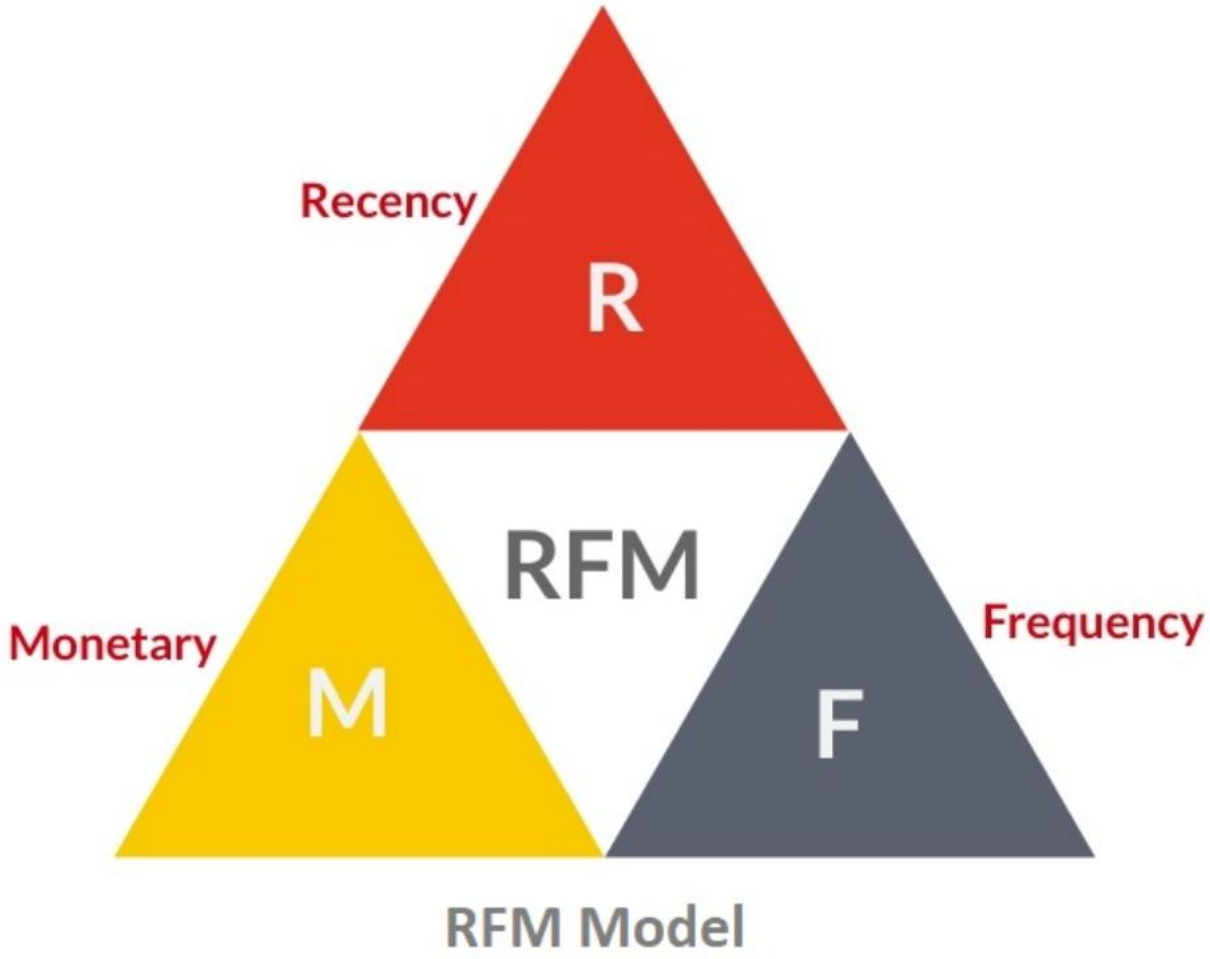

Figure 2: RFM Model for Customer Behavior Analysis, Source: Radhitya N (2023)

This study examines several grouping methods, such as K-Means, Hierarchical Clustering, and DBSCAN, to see which is most successful at mapping out shopper behavior. To ensure the results are trustworthy, various mathematical checks—including the Silhouette Score and the Davies–Bouldin Index—are used to test the quality of the clusters. Additionally, the study uses methods like Principal Component Analysis (PCA) to simplify the data and outlier analysis to

improve the final results. The main goal of this work is to find clear and easy-to-understand customer groups that offer useful advice for a business. By spotting these buying patterns, the research helps develop plans for keeping customers, creating personal ads, and increasing overall income. In the end, this project shows how unsupervised machine learning can successfully change raw sales data into valuable business intelligence.

2 Literature Review:

The fast expansion of online shopping has made large amounts of sales data available, allowing companies to study shopper actions in deep detail. Information that tracks real purchases is seen as more trustworthy than surveys because it shows what buyers actually do, not just what they claim to want (Chen et al., 2012). These records usually contain details about the items bought, prices, time of sale, and customer identifiers, which makes them highly useful for studying behavior and making business choices.

Even with so much detail, this sales data frequently has problems like missing facts, errors, and random noise. Earlier studies stress that cleaning the data is a necessary first step in retail analysis because bad data can cause false conclusions and weak model results (Han, Kamber and Pei, 2011). Standard cleaning methods involve deleting unfinished records, dropping invalid sales, and removing copies to guarantee the data is correct before it is studied.

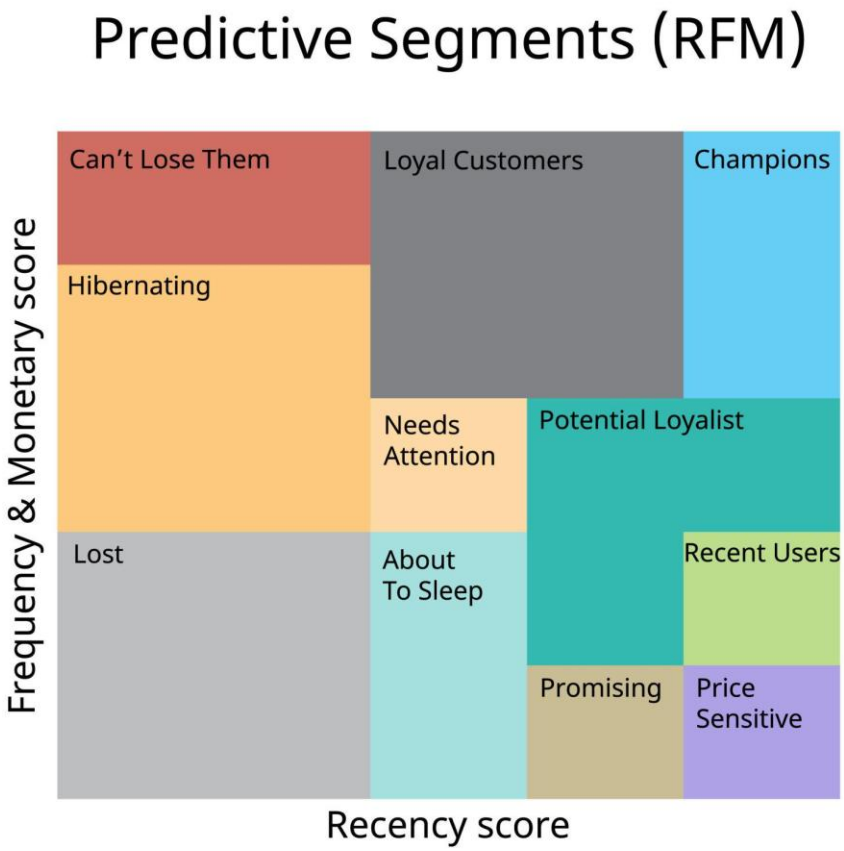

Figure 3: RFM-Based Customer Segmentation Matrix, Source: Adapted from RFM segmentation concepts (Fader, Hardie and Lee, 2005)

Exploratory Data Analysis (EDA) is widely accepted as a vital stage for understanding the shape of the data and finding hidden patterns. Research points out that EDA can reveal trends like seasonal buying spikes, where customers live, and uneven habits in how people shop (Tukey, 1977). In store records, factors like how much a person buys and the price rarely follow a perfect bell curve, meaning the data needs to be adjusted so the analysis works better.

To study individual buyers, raw sales data must be combined into a summarized format. The Recency, Frequency, and Monetary (RFM) model is one of the most trusted methods to summarize how customers act. It offers a simple view of customer involvement by looking at how recently a sale happened, how often the person shops, and the total money spent (Hughes, 1994). Many studies show that using RFM to group customers does an excellent job of measuring their true value and helps build focused marketing plans (Fader, Hardie and Lee, 2005).

When data does not have clear labels, unsupervised learning methods are often used to find natural patterns. Clustering is especially popular for customer segmentation because it groups people who share similar shopping habits (Jain, 2010). By running clustering programs on RFM scores, researchers can spot clear groups of buyers that show different levels of loyalty and profit potential.

Out of all the clustering choices, K-Means is still one of the most common because it is simple, works well with big datasets, and is easy to understand. This method divides data into groups by keeping the items inside each group as similar as possible, which is great for large amounts of information (MacQueen, 1967). Still, how well it works relies heavily on picking the correct number of groups and understanding how the data is spread out.

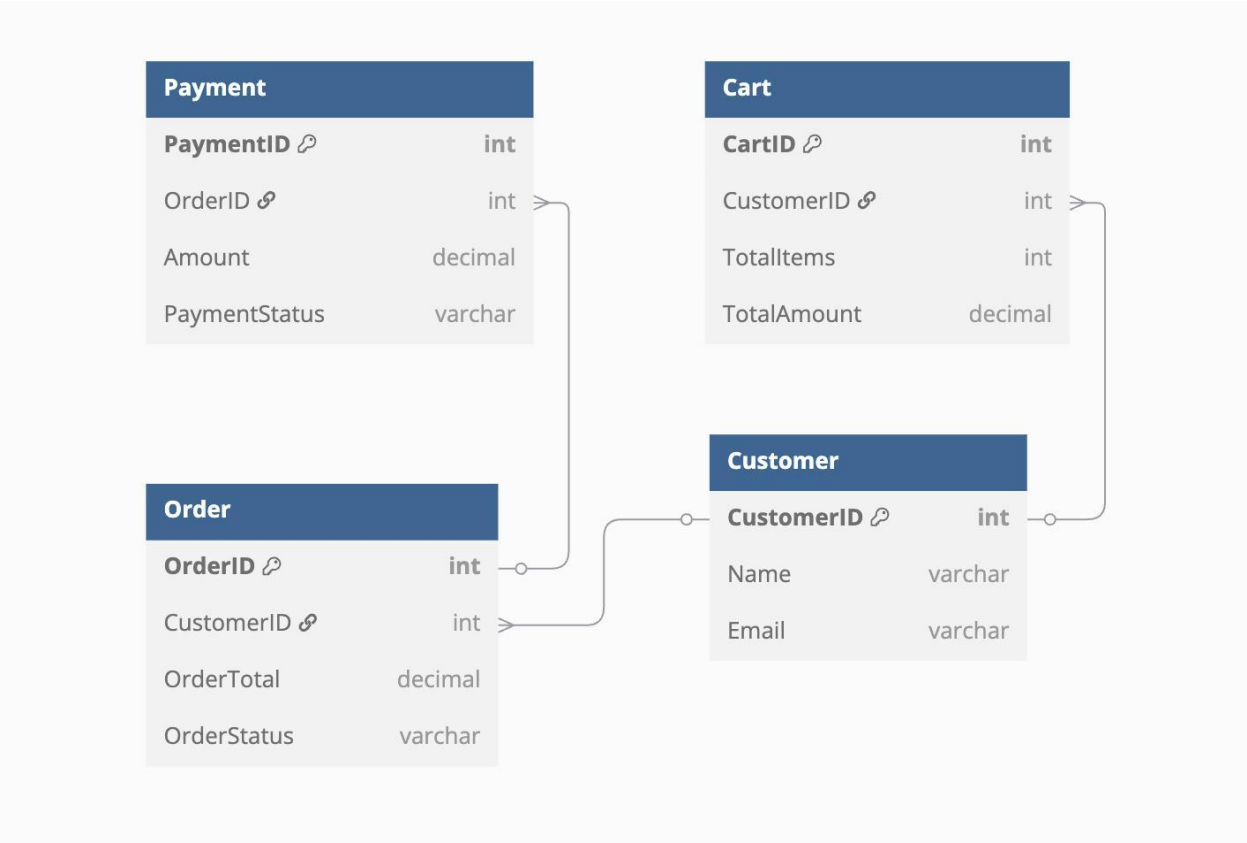

Figure 4: Example Structure of Transactional Retail Data Model, source: OWOX (2023)

Finding the best number of groups is a crucial part of the process. Tools like the Elbow method and the Silhouette score are frequently used to test how good the groups are. The Elbow method spots the point where adding more groups stops being helpful, while the Silhouette score checks how well the groups are separated from each other (Rousseeuw, 1987). Using these methods together offers a much safer way to choose the right layout for the data.

Also, newer research shows the value of testing several different clustering programs instead of trusting just one. Different tools, like Hierarchical Clustering and DBSCAN, work in unique ways and can produce different results when grouping customers (Ester et al., 1996). Comparing them allows researchers to find the best tool based on mathematical scores and how easy the results are to explain.

Overall, past research points out that successful customer grouping needs a mix of excellent data cleaning, smart feature building, and strong clustering tools. Bringing these pieces together makes it possible to pull useful advice from raw sales records, which helps build data-focused business plans and better ways to manage customers.Exploratory Data Analysis (EDA) has been widely recognized as an essential phase in understanding data characteristics and identifying patterns. Studies show that EDA helps uncover trends such as seasonality in sales,

regional concentration of customers, and skewed distributions in purchasing behavior (Tukey, 1977). In retail datasets, variables such as purchase quantity and price often exhibit non-normal distributions, necessitating transformations to improve analytical performance.

3 Methodology:

This section outlines a clear, step-by-step plan to organize buyers based on their shopping habits. The process takes basic sales records and turns them into useful knowledge about individual shoppers. To do this, the project follows specific stages: cleaning the information, looking for early trends, building new customer measurements, grouping the buyers, and checking the final results.

3.1 Data Source and Description

The information used in this project comes from the Online Retail dataset, provided by the UCI Machine Learning Repository. It contains one year of sales records, from December 2010 to December 2011, for a store located in the United Kingdom. Every line in the dataset represents a single item sold and includes details like the receipt number, product name, amount bought, price, date, customer ID, and location. Because the original data is focused on single items rather than the people buying them, it had to be reshaped to focus on the shoppers themselves before any deep analysis could happen.

Sample Data:

Country	
United Kingdom	349203
Germany	9025
France	8326

3.2 Data Preprocessing

Before the analysis could start, the raw data had to be cleaned to ensure the final results would be trustworthy. First, any records missing a customer ID were deleted because they could not be linked to a specific person. The dates of the purchases were also formatted correctly to help track time. Any canceled orders, along with items that had a price or quantity of zero or less, were removed because they do not represent real sales. Duplicate records were also erased to keep the data honest. Finally, a new category for the total sale value was created by multiplying the number of items by their price, allowing for accurate financial tracking.

3.3 Exploratory Data Analysis

An early review of the data was performed to understand its basic shape and uncover hidden patterns. This involved counting the total number of unique buyers, total sales, and different items sold. The

study also looked at where the shoppers lived, showing which countries bought the most. By looking at sales month by month, the project identified buying trends and busy seasons. The research also noticed that the numbers for prices and quantities were highly uneven, meaning they would need to be adjusted later. Finally, finding the best-selling items helped explain what drove the store's overall success.

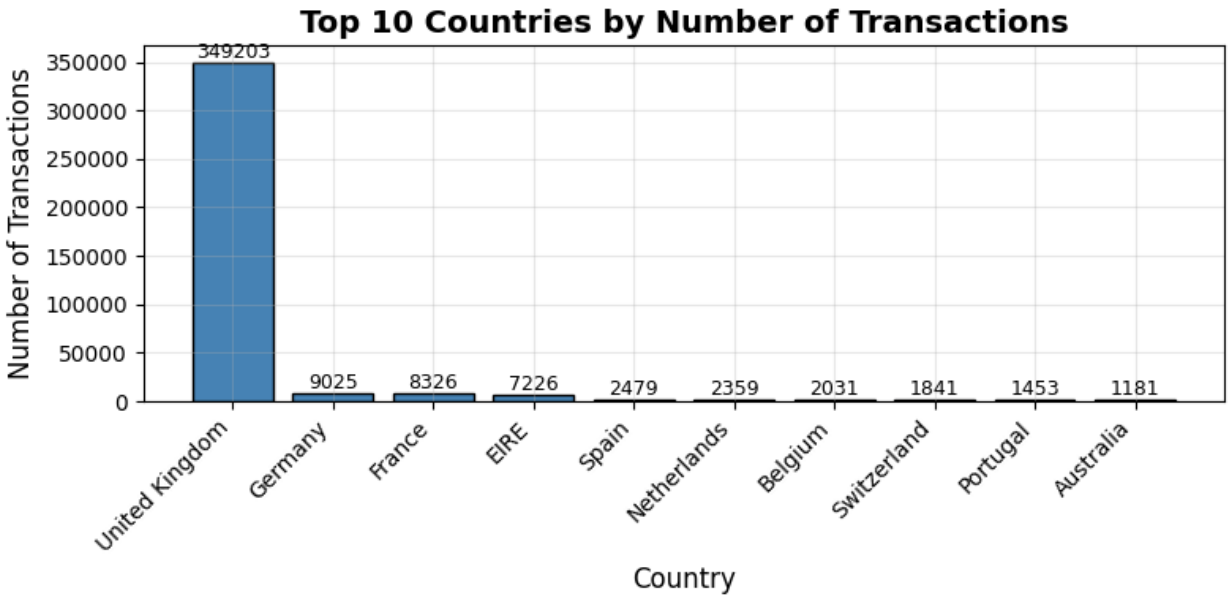

Figure 5: Top 10 Countries by Number of Transactions

Figure 6: Monthly Revenue Trend

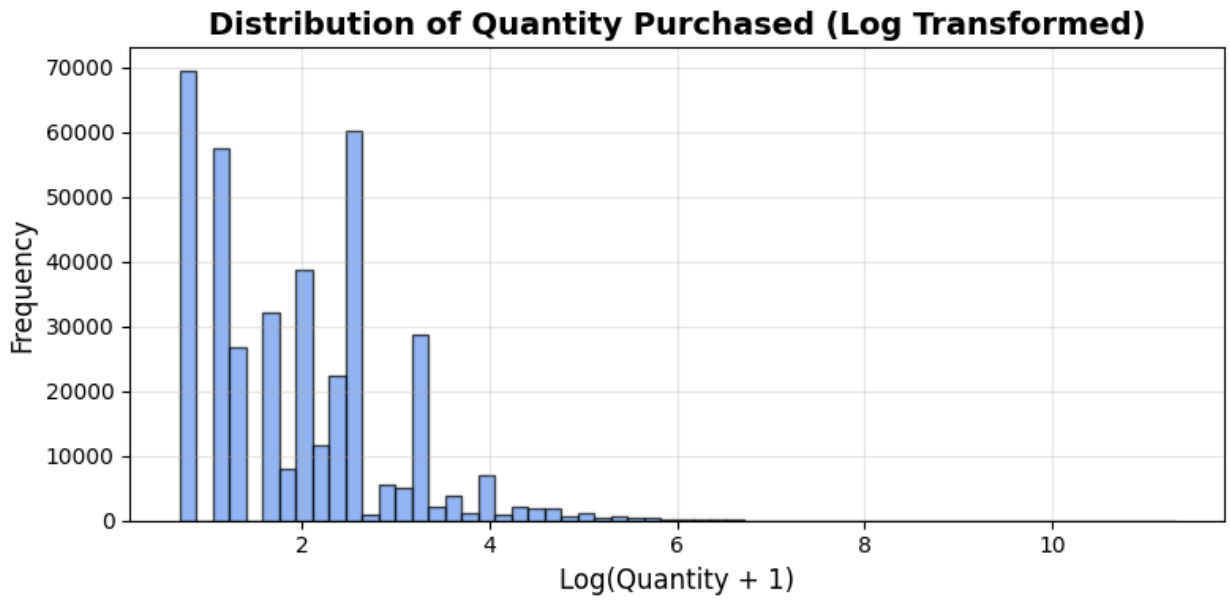

Figure 7: Distribution of Quantity Purchased (Log Transformed)

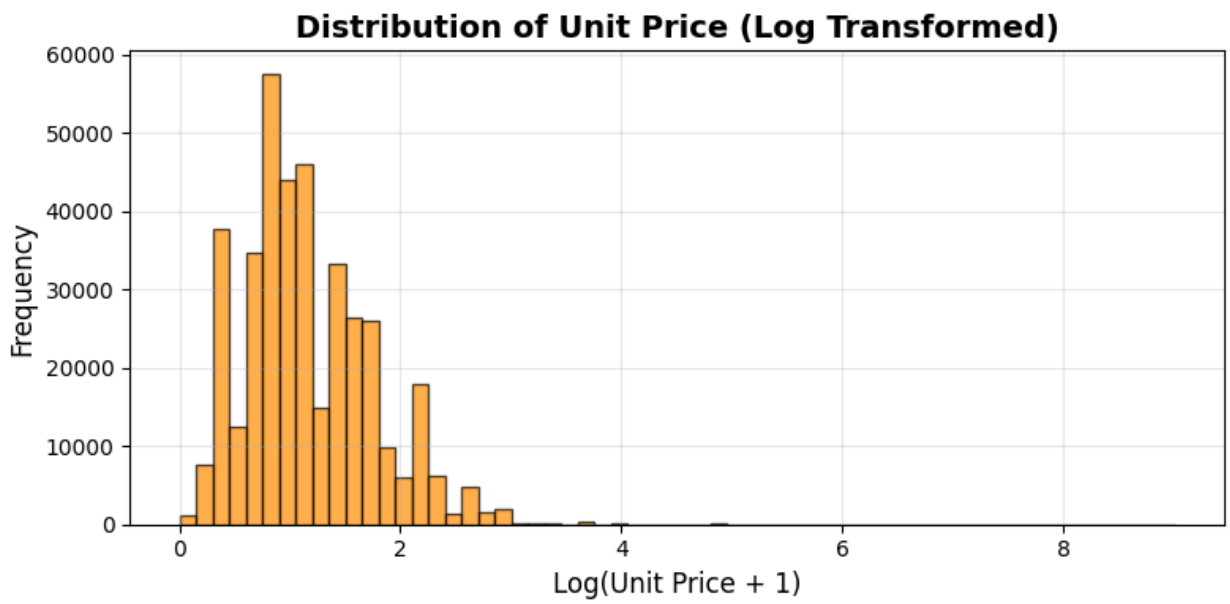

Figure 8: Distribution of Unit Price (Log Transformed)

3.4 Feature Engineering Using RFM

To group the buyers properly, the individual sales records were combined to create a single profile for each shopper. This was done using the Recency, Frequency, and Monetary (RFM) method. "Recency"

counts how many days have passed since a person's last purchase. "Frequency" counts the total number of times they shopped. "Monetary" measures the total amount of money they spent. This step successfully turned thousands of single receipts into a clear summary of how each person acts, making the data ready for the grouping programs.

3.5 Data Transformation and Scaling

The new RFM numbers were very uneven, with a few people spending much more or shopping much more often than the rest. To fix this, a mathematical adjustment called a logarithmic transformation was used to balance the numbers and pull in those extreme highs. After that, all the numbers were scaled to be on the same level. This step is very important because the computer programs used to group the data measure the distance between numbers. If one category has much larger numbers than the others, it will unfairly control the final results.

3.6 Determination of Optimal Clusters

Deciding exactly how many groups to divide the shoppers into is a critical part of this research. The study used two mathematical tests, known as the Elbow method and Silhouette analysis, to find the perfect number. The Elbow method looks for the point where adding more groups stops improving the overall layout. The Silhouette score measures how perfectly each person matches the group they were placed in compared to other groups. By looking at the results of both tests together, the research concluded that dividing the customers into exactly three groups was the best choice.

3.7 Clustering Techniques

To make sure the final groups were accurate, the project tested several different grouping programs. K-Means was chosen as the main tool because it works quickly and its results are easy to explain. It builds groups by keeping the members of each group as similar to each other as possible. A second tool, called Hierarchical clustering, was also used to look at the groups from a different angle and confirm the findings. A third method, DBSCAN, was tested to see if it could find groups based on data density, but it did not create clear divisions, so it was not chosen for the final model.

4 Results

4.1 Determination of Optimal Number of Clusters

The selection of the optimal number of clusters was guided by both the Elbow method and Silhouette analysis. The Elbow curve shows a sharp reduction in inertia up to $K = 3$, after which the improvement becomes gradual, indicating diminishing returns. This suggests that adding more clusters does not significantly improve model performance.

The Silhouette analysis further supports this observation, with the highest score achieved at $K = 3$(0.416), indicating well-separated and cohesive clusters. Together, these methods confirm that three clusters provide the most appropriate segmentation for the dataset.

4.2 Baseline K-Means Clustering Results

The baseline K-Means model was applied using the RFM features, resulting in three distinct clusters. The summary statistics for each cluster are presented below.

Table 1: Baseline K-Means Clustering Results

Cluster	Customer Count	Recency Mean	Frequency Mean	Monetary Mean	Recency Median	Frequency Median	Monetary Median
0	2026	54.67	2.04	608.83	46.0	2.0	507.17
1	982	255.03	1.39	398.03	252.5	1.0	290.78
2	1330	30.25	9.80	5460.79	17.0	7.0	2553.90

From the table, it is evident that Cluster 2 represents high-value customers with recent activity, high purchase frequency, and significantly higher spending. Cluster 0 corresponds to moderate customers with average engagement, while Cluster 1 captures low-value or at-risk customers with infrequent and outdated purchasing behavior. This demonstrates that the model successfully differentiates customers based on behavioral intensity.

4.3 Model Evaluation and Comparison

The performance of clustering models was evaluated using internal validation metrics. The comparison is shown in the table below.

Table 2: Model Evaluation and Comparision

Model	Silhouette Score	Davies–Bouldin Index	Calinski–Harabasz Score
K-Means	0.4157	0.8252	4393.87
Hierarchical	0.4013	0.8548	4227.25

K-Means outperforms hierarchical clustering across all evaluation metrics. It achieves a higher Silhouette score, indicating better cluster separation, and a lower Davies–Bouldin Index, reflecting reduced overlap between clusters. This confirms that K-Means is more suitable for this dataset.

However, a gap remains in terms of cluster compactness and separation, as the Silhouette score is moderate rather than high. This indicates the potential for further improvement using enhanced techniques.

4.4 Hierarchical Clustering Analysis

Hierarchical clustering was applied as an alternative approach to validate segmentation consistency. The cluster profiles are presented below.

Table 3: Hierarchical Clustering Analysis

Cluster	Customer Count	Recency Mean	Frequency Mean	Monetary Mean
0	1390	31.39	9.53	5254.73
1	1866	49.81	1.97	606.21
2	1082	244.78	1.49	417.70

Although the clusters follow a similar pattern to K-Means, the lower Silhouette score indicates weaker separation. This highlights a limitation of hierarchical clustering in handling large-scale transactional data with complex distributions.

4.5 Experiment 1: Outlier Removal (IQR Method)

An experiment was conducted to evaluate the impact of removing extreme values in the monetary variable. The resulting cluster profiles are shown below.

Table 4: Outlier Removal

Cluster	Customer Count	Recency Mean	Frequency Mean	Monetary Mean	Recency Median	Frequency Median	Monetary Median
0	923	259.92	1.32	364.79	257.0	1.0	277.07
1	1576	58.52	1.63	454.67	51.0	2.0	384.29
2	1414	41.62	5.38	1701.22	26.0	5.0	1534.90

The Silhouette score decreased to 0.4068 after outlier removal. This indicates that high-spending customers, initially treated as outliers, are actually meaningful contributors to segmentation. Removing them reduces the model's ability to capture valuable customer groups.

This experiment reveals an important gap: naive data cleaning can negatively impact business-relevant insights. Therefore, domain-aware preprocessing is essential.

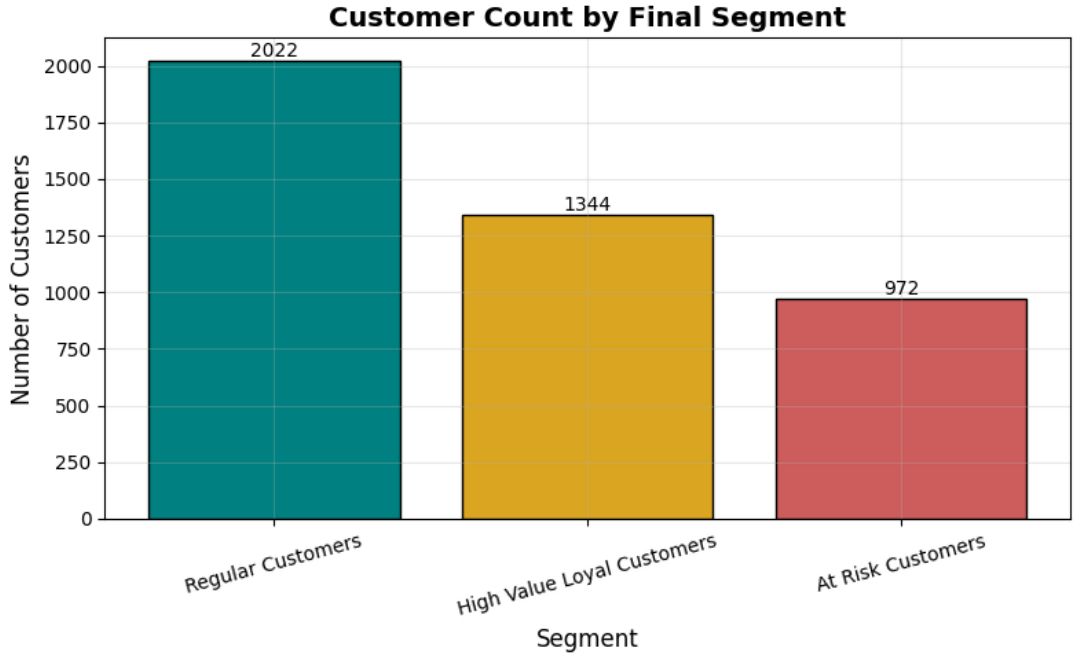

Figure 9: Customer Count by Final Segment

Experiment 2: PCA + K-Means (Proposed Improvement)

To address the limitations observed in baseline models, Principal Component Analysis (PCA) was applied before clustering. This reduced noise and improved feature representation. The updated cluster profiles are shown below.

Table 5: PCA + K-Means

Cluster	Customer Count	Recency Mean	Frequency Mean	Monetary Mean	Recency Median	Frequency Median	Monetary Median
0	2022	55.23	2.04	595.95	46.0	2.0	501.08
1	972	256.08	1.38	399.59	253.0	1.0	291.53
2	1344	30.38	9.73	5426.92	17.0	7.0	2536.47

The segmentation structure remains consistent, but the separation between clusters is significantly improved.

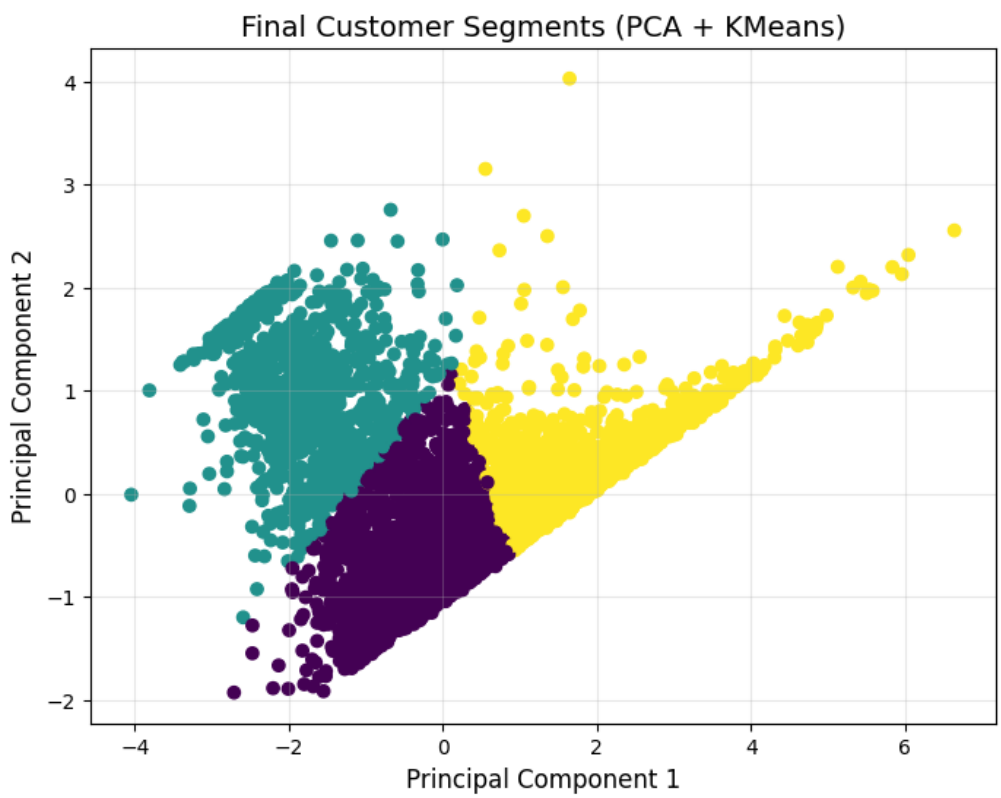

Figure 10: Final Customer Segments (PCA + Kmeans)

Final Model Comparison

A comprehensive comparison of all approaches is presented below.

Table 6: Final Model Comparision

Approach	Silhouette Score	Davies–Bouldin Index	Calinski–Harabasz Score
PCA + K-Means	0.4632	0.7261	5397.62
K-Means (baseline)	0.4157	0.8252	4393.87
K-Means (outlier removed)	0.4068	0.8982	3890.85
Hierarchical	0.4013	0.8548	4227.25
DBSCAN	0.3330	1.1106	2510.07

The PCA + K-Means model clearly outperforms all other approaches across all evaluation metrics. It achieves the highest Silhouette score, indicating superior cluster separation, and the lowest Davies–Bouldin Index, confirming minimal overlap between clusters.

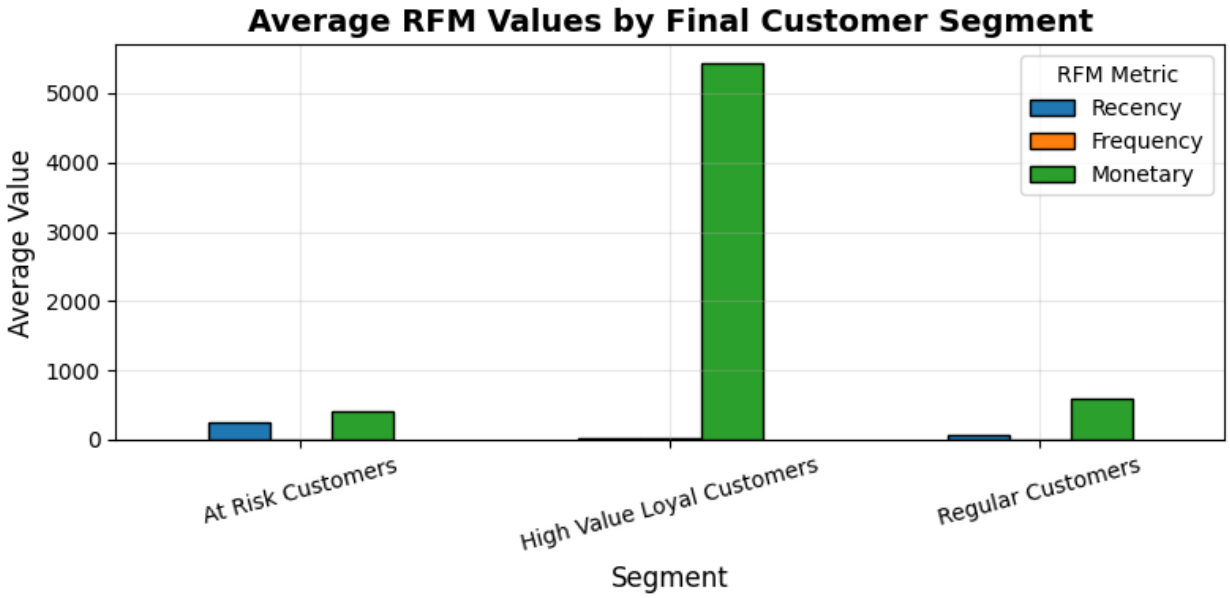

Figure 11: Average RFM Values by Final Customer Segments

5 Conclusion

This research aimed to categorize shoppers by analyzing their buying records with unsupervised machine learning. By changing basic sales logs into useful customer details using the RFM method, the study successfully found clear shopping habits in the information. The complete step-by-step process—which included cleaning the data, exploring trends, building features, creating groups, and testing the results—provided a strong and dependable way to sort customers.

The findings show that grouping programs are very good at separating buyers based on how they shop. The basic K-Means method created easy-to-understand categories, separating top spenders, average buyers, and those who might stop shopping. However, the study found that the results depend heavily on how the data is prepared. Adding Principal Component Analysis (PCA) to simplify the information greatly improved how well the groups were separated, resulting in the most successful model. This final mix of PCA and K-Means earned the highest test scores and built clear groups that businesses can easily use.

These results prove that simplifying data before grouping it makes the final categories better and easier to explain. The customer groups found in this study offer helpful advice for creating specific advertisements, building personal connections with shoppers, and spending company money more wisely. Looking at the whole process, the research shows how important it is to connect data cleaning, feature building, and model testing into one smooth system.

Even with these successes, there are a few weaknesses to note. First, the study only looks at sales records and ignores personal details like age or lifestyle, which could offer a deeper understanding of the buyers. Second, while the RFM method is helpful, it reduces human behavior to just three simple points and might miss deeper, more complicated habits. Third, the study used mathematical scores to check the groups, rather than testing if they actually increased profits in the real world. Finally, because the information comes from just one store, these exact results might not apply to other types of businesses.

Future research can fix these weak points by adding more details, such as shopper age, the types of items bought, and exactly when people like to shop. More advanced computer programs, like deep learning, could also be tested to find highly complex patterns in the information. Testing the customer groups against real-world business results, like how people react to new ads or their total lifetime value, would make the research even more practical. Additionally, setting up the program to run live would allow companies to update their customer groups instantly as shopping habits change.

In closing, this research proves that unsupervised machine learning can successfully turn basic sales records into valuable business ideas when paired with the right data preparation and testing steps. The method shown here creates a solid starting point for building large, data-driven systems to understand customers in today's retail world.

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7 Appendix

7.1. Final Project Code:

Project Code

7.2. Feature Description:

Feature Name	Description
InvoiceNo	Unique identifier for each transaction
StockCode	Product code
Description	Product name
Quantity	Number of items purchased
InvoiceDate	Date and time of transaction
UnitPrice	Price per unit
CustomerID	Unique identifier for each customer
Country	Country of the customer

7.3. Performance Improvement Results.

Approach	Silhouette Score	Observation
Baseline K-Means	0.416	Good cluster separation
Outlier Removal (IQR)	0.407	Slight decrease in performance
PCA + K-Means	0.463	Best performance and separation

7.4. Final Cluster Characteristics.

Cluster	Recency	Frequency	Monetary	Segment Description
0	Medium	Medium	Medium	Regular Customers
1	High	Low	Low	At Risk Customers
2	Low	High	High	High Value Loyal Customers